

Streamlined, Reproducible Data Analysis with NF-Core

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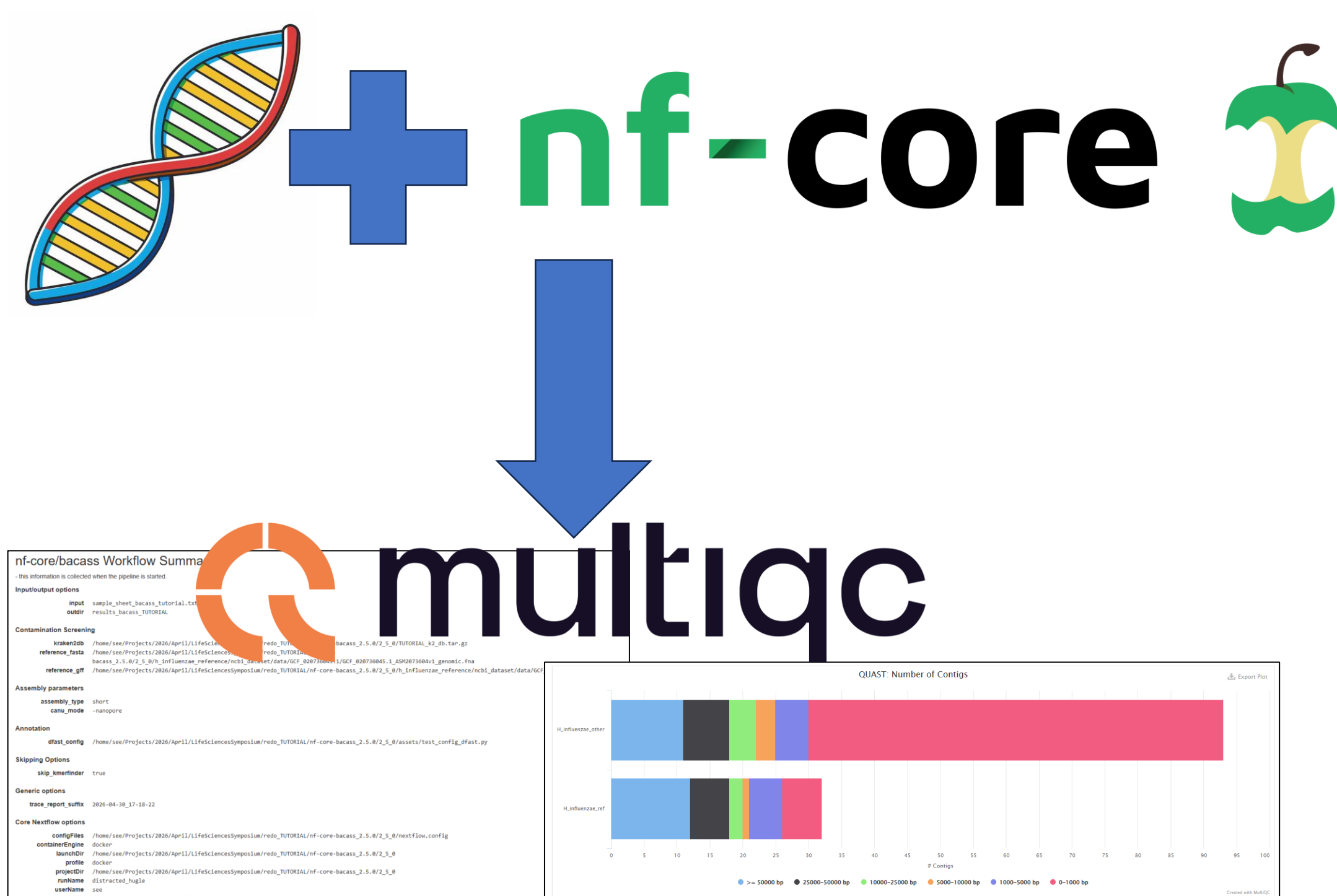
Introduction

- Reproducibility is vital to science
- However, many say science is experiencing a reproducibility crisis¹
- Reproducibility for computational life sciences may be as low as 20%¹
- Guidelines exist (e.g. FAIR), but guidelines alone are not enough²
- And, thus the nf-core community was born²
- Its goal is to facilitate analysis with best-practices for reproducibility²
- Accordingly, nf-core has a multitude of publicly available tools²

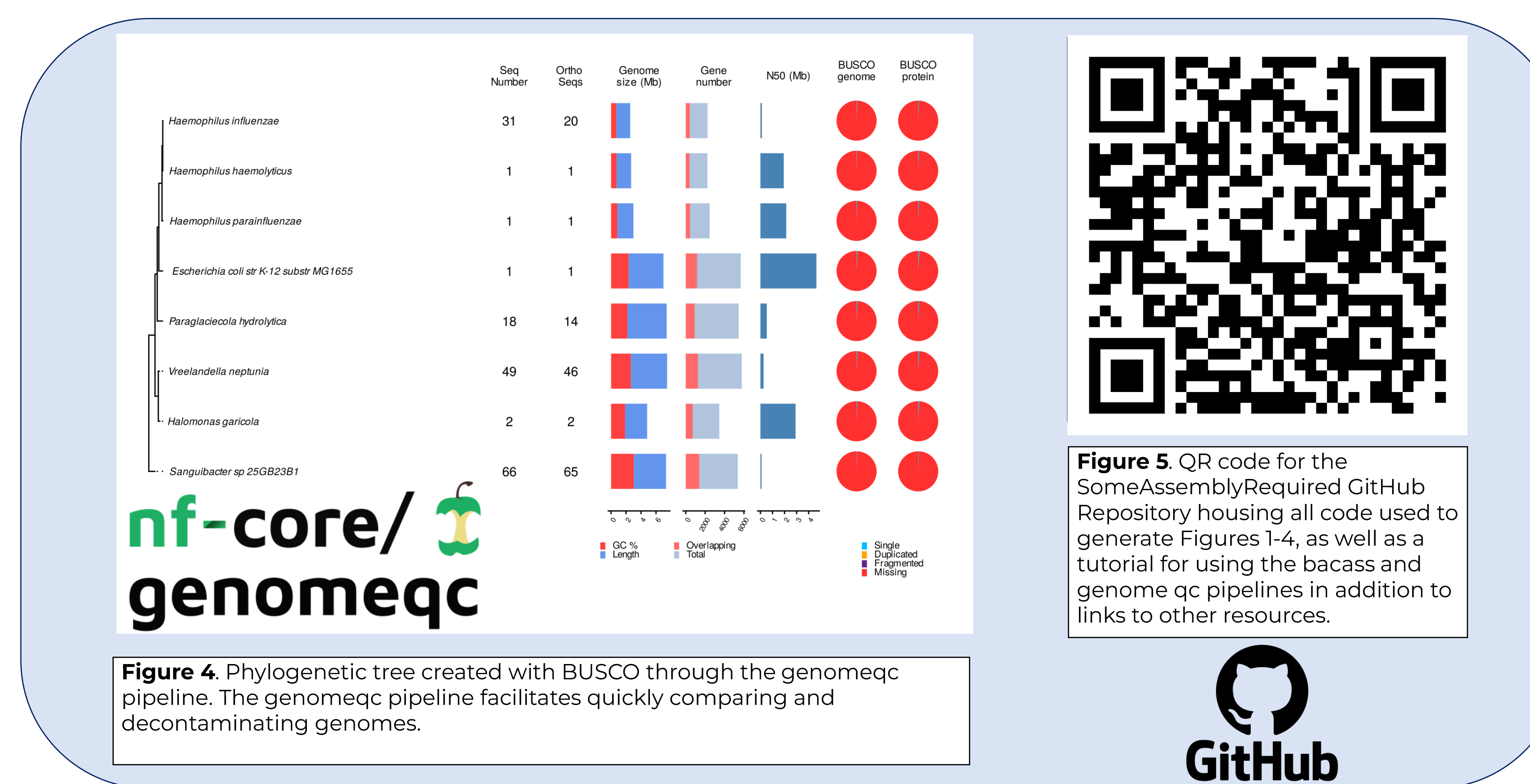
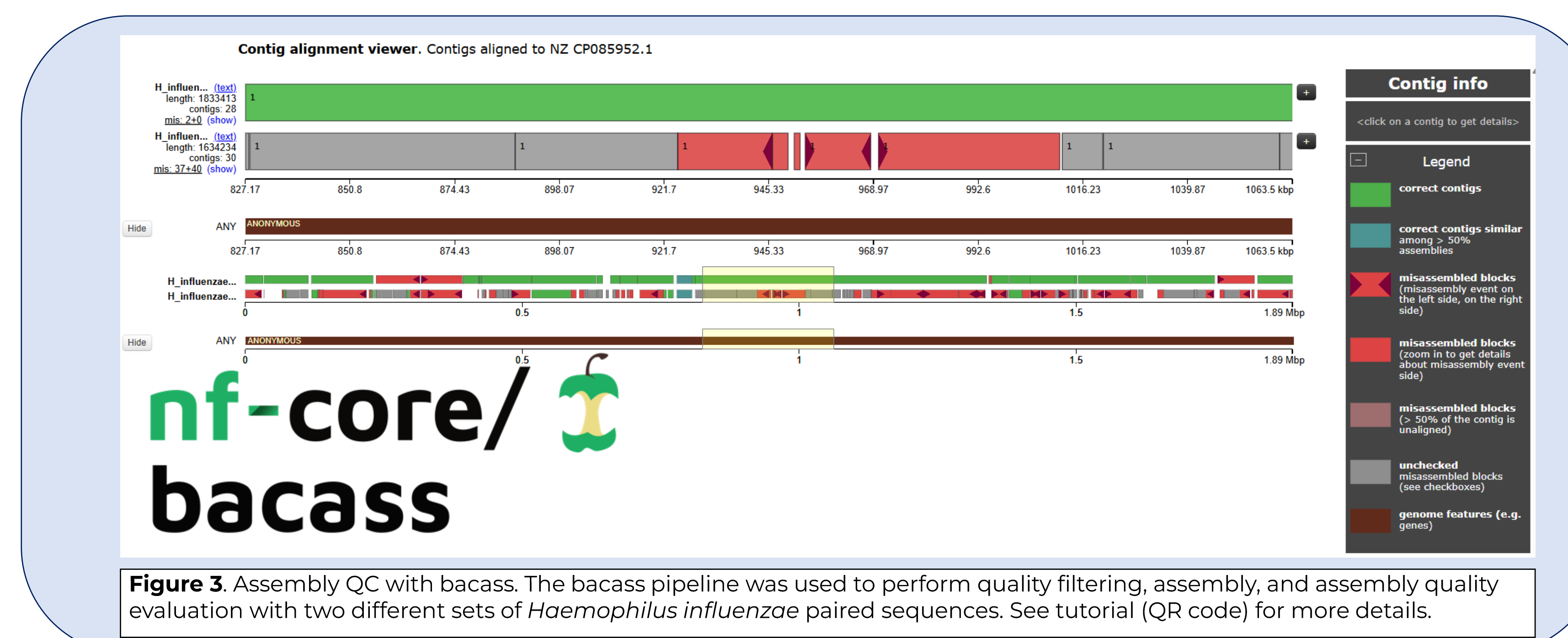
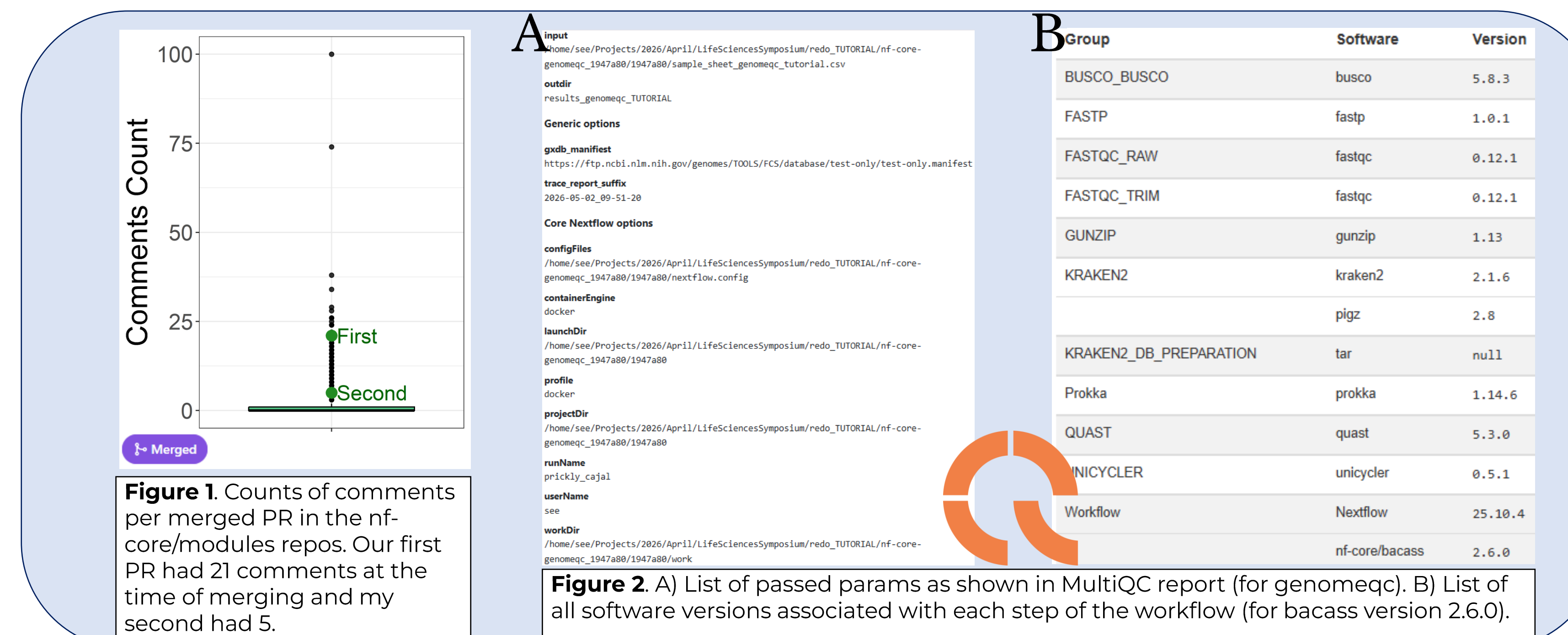
Objectives

- Learn about nf-core resources
- Make a tutorial showcasing them

Methods



Results



Discussion

- Writing reusable code can be more time-consuming but using it is faster
- We can take advantage of existing publicly available resources
- The nf-core community has released 91 computational pipelines
- Responsibility is still on the user to make sure all analyses are appropriate
- All nf-core pipelines track software versions and parameters
- Reproducibility is always important, but it does NOT have to be hard

Future Directions

- Improve the GitHub tutorial
 - Update GenomeQC version
 - Add more screenshots
- Integrate tutorial into an existing class

References

1. Ziemann, M., Poulain, P. & Bora, A. The five pillars of computational reproducibility: bioinformatics and beyond. *Brief. Bioinform.* **24**, bbad375 (2023).
2. Ewels, P. A. et al. The nf-core framework for community-curated bioinformatics pipelines. *Nat. Biotechnol.* **38**, 276–278 (2020).
3. VM, D. et al. nf-core/bacass: nf-core/bacass v2.6.0 - Crimson Titanium Seahorse. Zenodo <https://doi.org/10.5281/zenodo.20036081> (2026).
4. FernandoDuarteF et al. nf-core/genomeqc. nf-core (2026).

Acknowledgements

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